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A Paleoproterozoic giant radiating dyke swarm in the Dharwar Craton, southern India

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[1] Identical high precision U-Pb baddeleyite ages, together with paleomagnetic and geochemical data, on mafic dykes occurring over an area of 140,000 km^2, define a Paleoproterozoic giant dyke swarm at ca. 2.367 Ga in the Dharwar craton, south India, referred to here as the Dharwar giant dyke swarm. All six U-Pb ages on these dykes are identical within error and suggest emplacement of this swarm within a geologically short time span of ~5 Myr. A systematic southward progression in the trend of dykes from N48°E to N90°E, defines a fan angle of about 40° with convergence to a focal point about 300 km west of the present-day Dharwar craton boundary, resulting in a spectacular radiating dyke swarm extending across the entire eastern Dharwar craton. The large areal extent, radiating dyke pattern and short duration imply a mantle plume origin for the Dharwar giant dyke swarm. Despite their large areal distribution, all dykes in this swarm are geochemically coherent and have similar primitive mantle-normalized trace element patterns and rare earth element characteristics. Although the NE part of the swarm is magnetically overprinted, a remanence survives that has the same direction as primary magnetizations from dykes in the southern part of the swarm.

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1. Introduction

[2] In many Large Igneous Provinces, mafic dyke swarms frequently represent a major component of the magmatic conduit system. Dykes tend to become more exposed with time as shallower expressions of magmatism such as lava flows are removed by erosion. Dyke swarms typically constitute hallmark representatives of mafic igneous activity associated with major continental rifting/plume events [*Halls and Fahrig*, 1987; *Ernst and Buchan*, 1997]. Their excellent preservation, worldwide occurrence both in space and time, and ability to retain high quality U-Pb geochronological, paleomagnetic and geochemical data make them ideal subjects for reconstructing past continental configurations and for addressing the concept [e.g., *Li and Zhong*, 2009] that Earth's history has been characterized by repeated supercontinent formation and disaggregation. In this paper we seek to define a giant radiating dyke swarm, the scale of which leads to the expectation of its presence on other continents that may have been subjacent to India in the early Proterozoic, thereby helping to provide a critical element to continental re-assembly at this time.

[3] The first map of dyke swarms from the Dharwar craton, India was based on Landsat data [see *Halls*, 1982, Figure 2]. At that time, the speculation was made [see *Halls*, 1982, Figure 2] that a large radiating dyke swarm crossed the Dharwar craton and supported an earlier observation [*Halls*, 1978] that the earliest-formed dyke swarms subsequent to cratonization appear to be everywhere normal to the older structural grain of the Archean basement. A Paleoproterozoic age for this Indian dyke swarm was based primarily on a Rb-Sr date [*Ikramuddin and Stueber*, 1976] but no supportive U-Pb geochronology, paleomagnetism or geochemistry were available to show whether or not the dyke swarm was a single magmatic event. The southern part of the radiating dyke pattern, known as the Bangalore swarm [*Halls et al.*, 2007a], is formed of early Proterozoic dykes with U-Pb baddeleyite ages of 2365–2369 Ma [*Halls et al.*, 2007b; *French and Heaman*, 2010]. The northern part of the radiating pattern is formed by the hitherto undated Karimnagar dyke swarm [*Rao et al.*, 1990]. Paleomagnetic directions from these dykes were measured by *Rao et al.* [1990], and are almost 90° different from those found in Bangalore dykes [*Halls et al.*, 2007b] suggesting that the two swarms are not the same age. This paper presents a combined U-Pb geochronology, paleomagnetic and geochemical study of the Karimnagar dykes in order to fully establish whether or not a single early Proterozoic giant radiating dyke swarm exists across the entire Dharwar craton.

2. Background Geology

[4] Most of the southern peninsula of India is underlain by the Archean Dharwar Craton. It is blanketed by the Cretaceous Deccan basalts in the north and northwest and is restricted by the Godavari rift to the northeast (Figure 1). On the east, the craton is bounded by the ~1.5 Ga Eastern Ghats mobile belt and on the west by the Arabian Sea. The Archean crust as summarized by *Swami Nath and Ramakrishnan* [1981], *Drury et al.* [1984] and *Naqvi and Rogers* [1987] consists of a southern granulite block and a low-grade northern block. The northern block has been further divided into a low temperature western and a high temperature eastern block (based on contrasting metamorphic *P-T* conditions), separated by a N-NW trending fault zone. Amalgamation of the western and eastern Dharwar cratonic blocks is interpreted to have taken place at ~2.5 Ga [*Chadwick et al.*, 2000; *Balakrishnan et al.*, 1990].

[5] As in many Archean crustal blocks, the Dharwar craton is transected by large numbers of mafic dykes. Here, they define swarms which are predominantly of Paleoproterozoic age [*Halls et al.*, 2007b; *French et al.*, 2008; *French and Heaman*, 2010, and references therein], with occasional younger dykes of Cretaceous age [*Kumar et al.*, 1988; *Radhakrishna et al.*, 1994]. Of these, the ESE-WNW to E-W Bangalore dyke swarm, covering most of the south and central parts of Dharwar craton [*Halls et al.*, 2007b], appears to be the most dominant. Four high precision U-Pb baddeleyite ages have been reported from these dykes (Figure 1): at Henur: 2366.7 ± 1.0 Ma [*Halls et al.*, 2007b], Harohalli: 2365.4 ± 1.0 Ma, Penukonda: 2365.9 ± 1.5 Ma and Chennekottapalle: 2368.6 ± 1.3 Ma [*French and Heaman*, 2010], and suggest that the Bangalore swarm was emplaced within a geologically short time span of ~5 Myr or less.

[6] The southern dykes are characterized by a steep negative remanence to the SE [*Kumar and Bhalla*, 1983; *Radhakrishna and Joseph*, 1996; *Halls et al.*, 2007b]. This remanence is thought to be primary because the Penukonda dyke yields a positive baked contact test with an older dyke [*Kumar and Bhalla*, 1983]. The northern, NE-trending dykes, as represented by the Karimnagar dyke swarm,

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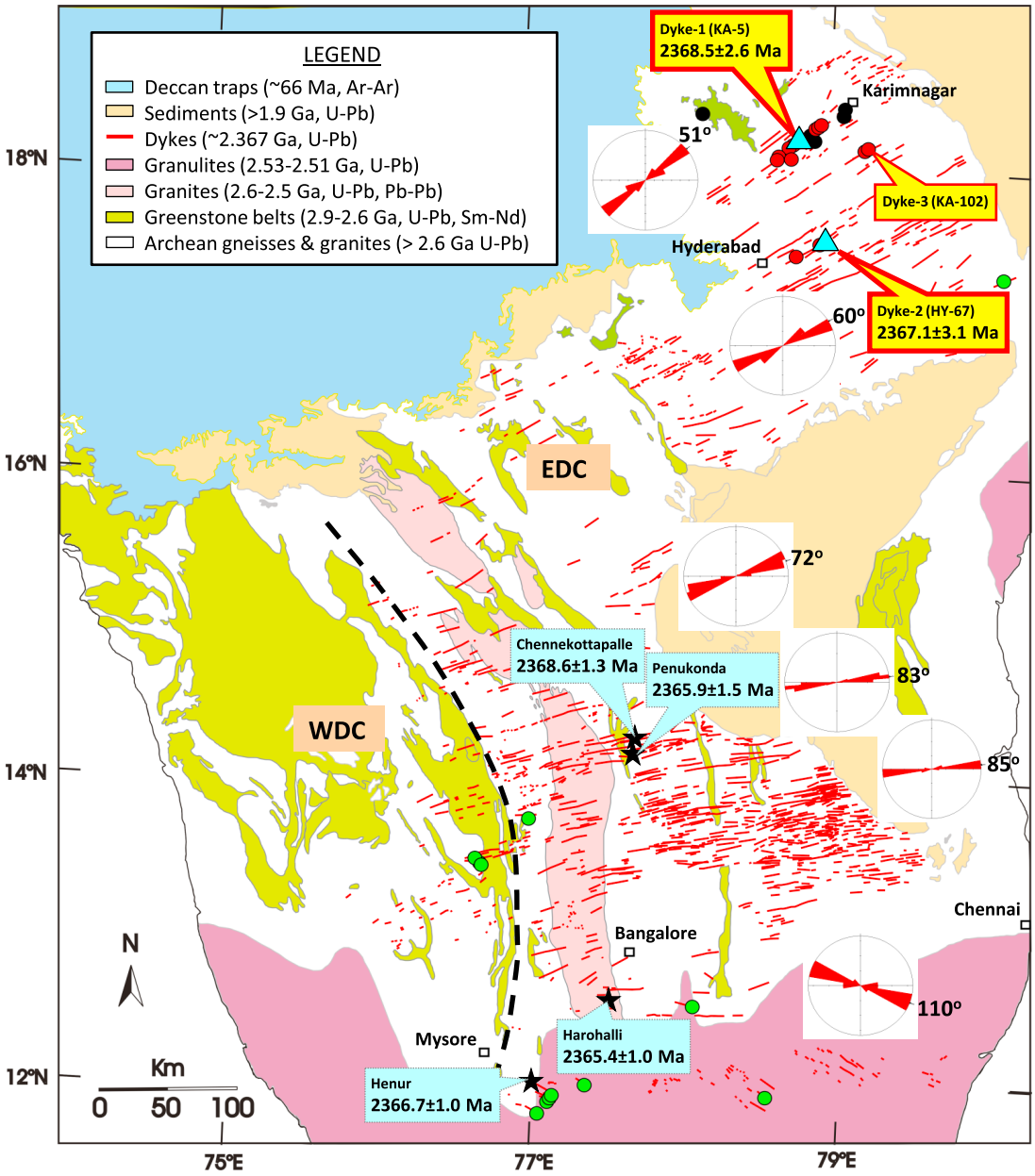


Figure 1. Simplified geology of south India (modified after *Geological Survey of India* [1998] and *Naqvi and Rogers* [1987]), showing the distribution of ca. 2367 Ma dykes (dyke distribution after *French and Heaman* [2010], *Halls et al.* [2007b] and Google Earth maps). EDC and WDC are Eastern Dharwar Craton and Western Dharwar Craton, respectively. Black dashed line demarcates the boundary between the two blocks. Blue triangles on the map indicate site locations of samples used for U-Pb age determinations in the present study and black stars in earlier studies. Red and black dots are sampling sites in the Karimnagar swarm representing present and earlier studies [*Rao et al.*, 1990], respectively, and green dots represent site locations in the Bangalore dyke swarm region (paleomagnetic and geochemical data for these sites were reported earlier by *Halls et al.* [2007b]) for which additional geochemical data is added in the present study. Regional strike variation within the dyke swarm from north to south is shown in the form of rose diagrams.

have yielded a direction with a shallow negative inclination and NE declination [*Rao et al.*, 1990]. However, the within-site scatter is unusually large, and host rock charnockites remote from Karimnagar dykes at the northern end of the swarm give a similar direction [*Bhimasankaram*, 1964], raising suspicion that the magnetization is an overprint. Geochemical data have been reported from both the Bangalore [*Halls et al.*, 2007b; *French and Heaman*, 2010] and Karimnagar dykes

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[*Rao et al.*, 1990]. Both sets of data indicated that the dykes are sub-alkaline tholeiites, but were sufficiently incomplete to prevent a definitive case being made that both groups of mafic dykes constitute part of the same swarm.

3. Sampling Strategy

[7] NE trending dykes of the Karimnagar swarm were sampled at 23 locations in the vicinity of Hyderabad and Karimnagar (Figure 1) for U-Pb geochronology, paleomagnetism and whole rock geochemistry (on samples collected within about 50 cm of dyke margins). All sites are located over 400 km north of the previously reported results on Bangalore dykes (Figure 1). The methodologies used are described briefly in the auxiliary material.[1]

4. Results

4.1. Petrography and Geochemistry of Bangalore and Karimnagar Dykes

[8] The Karimnagar and Bangalore dykes are all mineralogically similar, consisting primarily of plagioclase and pyroxene, in an ophitic to sub-ophitic texture, with minor amounts of amphibole, myrmekite, biotite and magnetite and an absence of olivine. In dykes from the NE end of the Karimnagar swarm and from the southern parts of the Bangalore swarm, where host rocks have a higher metamorphic grade, feldspars are suffused with tiny inclusions causing a cloudy appearance [*Halls et al.*, 2007b; *Rao et al.*, 1990].

[9] Full major, trace and rare earth element compositions were obtained on dyke samples from both the Karimnagar and Bangalore regions (see table in Text S1). All the investigated dykes exhibit limited variation in major element concentrations and plot along a tholeiitic trend on a ternary AFM diagram (Figure 2a) [after *Irvine and Baragar*, 1971], and due to their moderate SiO_2 (50.7 to 54.3 wt %) and low total alkali contents ($Na_2O + K_2O$ = 1.78 to 2.86%) are classified chemically as basalts and basaltic andesites (Figure 2b) [*Le Maitre et al.*, 2002]. The most noticeable major element geochemical discriminant between the different dyke groups is a distinctly higher MgO content in Karimnagar dykes (ca. 7.4–8.6 wt. %) compared to that in Bangalore dykes (ca. 4.6–6.0 wt. %), suggesting that the latter formed from a more evolved batch of magma. Primitive mantle-normalized trace element variations (Figure 3a) reflect this evolution, showing generally parallel patterns and smooth trends with overall increasing levels of trace element enrichment in concert with increasing degree of differentiation (e.g., decreasing *mg#*), with overlapping Zr/Nb (Karimnagar: 17.0 to 20.6 and Bangalore: 13.5 to 23.9) and La/Nb (Karimnagar: 2.37 to 3.31 and Bangalore: 1.51 to 2.49) ratios. A chondrite-normalized rare earth element (REE) plot (Figure 3b) shows parallel patterns for samples across the swarm with relatively constant ratios of $[La/Sm]_N$, mostly between 1.75 to 2.33, $[Gd/Yb]_N$ of 0.97 to 1.20, and $[La/Yb]_N$ from 1.78 to 3.33 (chondrite normalizing values of *McDonough and Sun* [1995]), suggesting geochemical coherence across the Dharwar giant dyke swarm.

[10] Ratios of immobile incompatible trace elements can serve as useful indicators for fingerprinting petrogenetic processes and magmatic provenance by seeing through some aspects of igneous fractionation (e.g., melting, crystallization) in basaltic systems, including mafic dyke swarms [e.g., *Fitton et al.*, 1997; *Condie*, 1997]. As noted by *Pearce* [2008], ratio-ratio plots such as Th/Yb versus Nb/Yb can be effective for assessing the relative roles of source variation versus contamination in the evolution of mantle-derived basic magmas. Figure 4 illustrates such relationships for the Bangalore and Karimnagar dykes. The data for both subsets of the Dharwar giant dyke swarm are displaced vertically above the MORB-OIB array [*Pearce*, 2008]. These compositions trend sympathetically to high Th/Nb ratios (0.23–0.73; table in Text S1) together with a relatively steep, paired dispersion to high Th/Yb ratios, toward various estimates of the composition of average continental crust. Low initial concentrations of Th and Nb in the parental magma(s) (appropriate for compositions near that for primordial mantle (PM) or N-MORB; Figure 4) have likely made them sensitive to coupled crust-mantle interaction processes such as assimilation-fractional crystallization, rendering them well-displaced from the MORB-OIB array [*Pearce*, 2008].

[11] Such trends for these dykes are reproduced in other (immobile) incompatible element ratio plots such as Th/Nb versus Zr/Y, and Th/Ta versus La/Yb [e.g., *Condie*, 1997]. Tendencies toward higher Th/Nb and La/Yb ratios in the Karimnagar dykes (which tend also to have higher SiO_2) suggest that these may show a slightly greater degree of contamination, even though they appear to be less

[1]Auxiliary materials are available in the HTML. doi:10.1029/2011GC003926.

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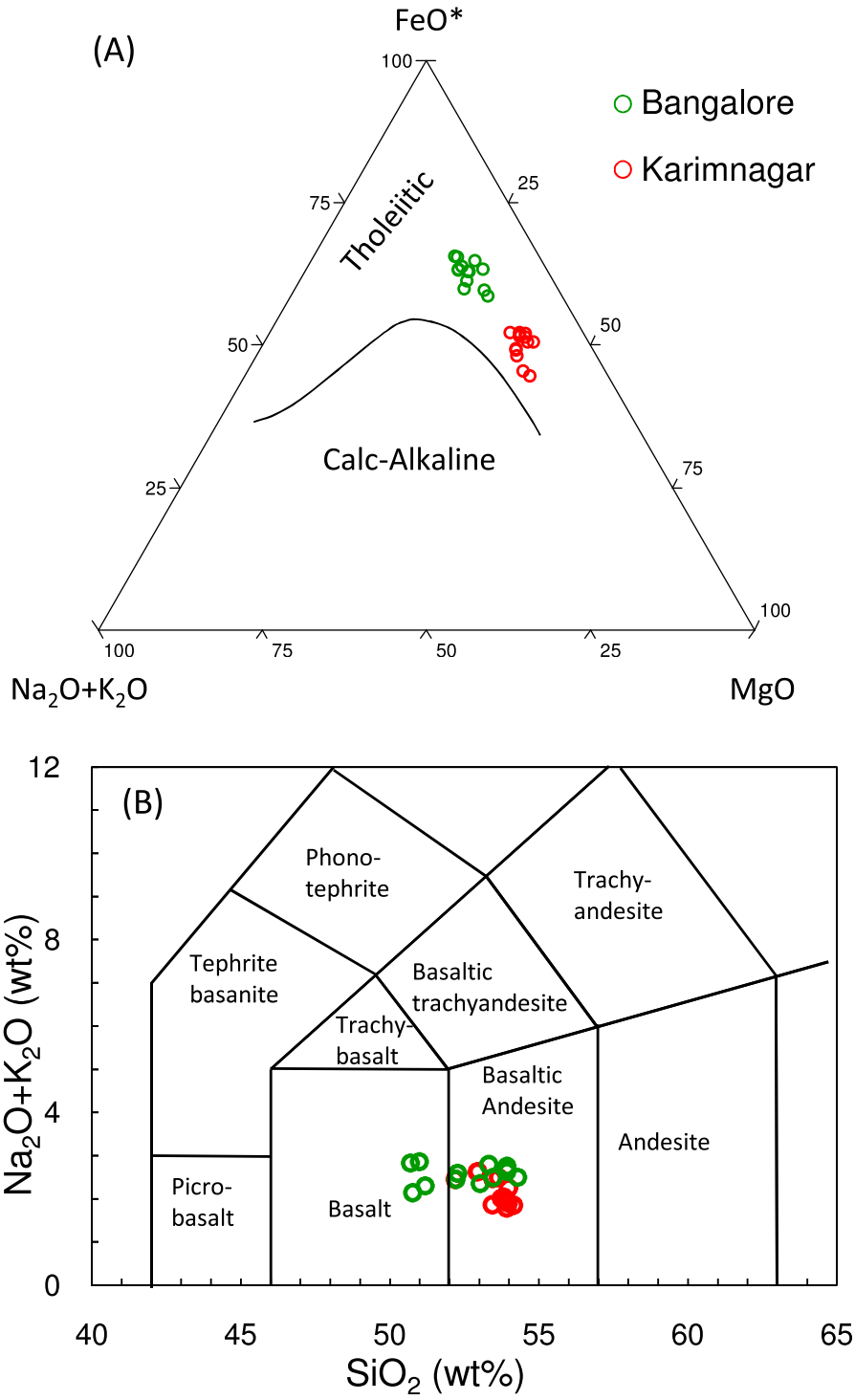


Figure 2. (a) AFM diagram and (b) total alkali versus silica diagram showing the compositions of dykes from both the Karimnagar and Bangalore swarms for comparison. Symbols for the Karimnagar and Bangalore dykes are the same in both figures. See Figure 1 for sample locations.

evolved (higher mg#s). This could also reflect, in part, a difference in the crustal contaminant in this part of the Dharwar craton, or some variability in source region or parental magma composition.

4.2. U–Pb Geochronology

[12] Results of U–Pb ID-TIMS (isotope dilution thermal ionization mass spectrometry) analyses of

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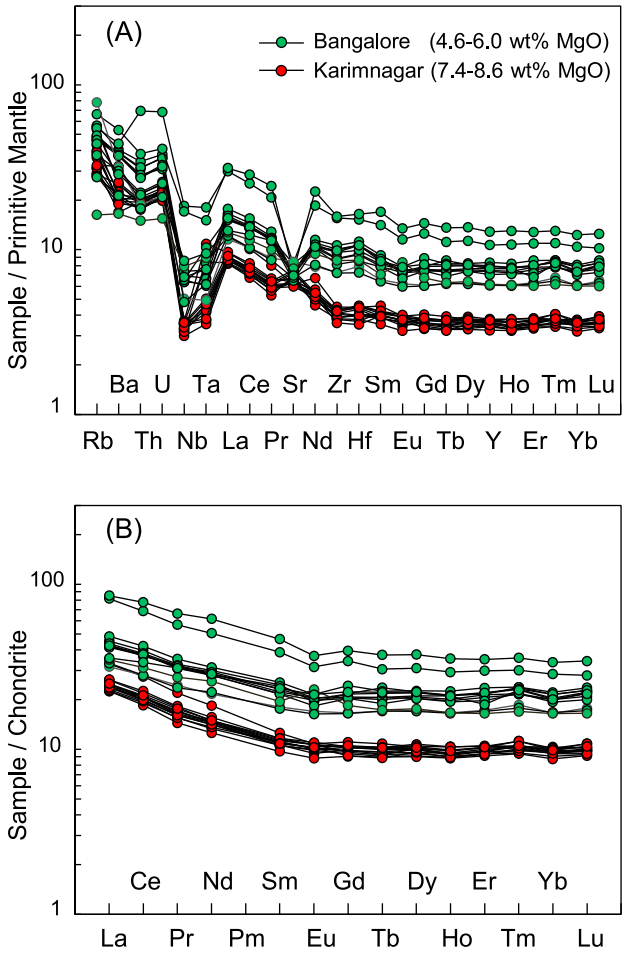


Figure 3. (a) Primitive mantle-normalized multielement diagram from selected dykes. (b) Chondrite-normalized REE diagram for the same set of samples. Normalizing values taken from *McDonough and Sun* [1995].

rare and small (<50 μm) crystals of baddeleyite (ZrO_2) recovered from two NE-trending dykes, one near Karimnagar (dyke 1) and one near Hyderabad (dyke 2) are presented in Figures 5a and 5b, respectively (U-Pb data are provided in Table 1). Data for dyke 1 (sample KA-5), yield a well-constrained upper intercept age of 2368.5 ± 2.6 Ma, while that for dyke 2 (sample HY-67) gives a similar intercept age of 2367.1 ± 3.1 Ma. These new ages fall within the narrow age span of the four Bangalore dykes (2365.4 ± 1.0 Ma to 2368.6 ± 1.3 Ma [*Halls et al.*, 2007b; *French and Heaman*, 2010]). The results dramatically show that the Bangalore and Karimnagar dykes are coeval, so that a single, craton-scale radiating swarm, emplaced within an exceptionally brief time span of approximately 3 Myr (maximum error limits permit ~7 Myr), can now be defined (see Figure 1).

[13] Linear regressions of data from both samples yield Neoproterozoic lower intercept ages (ca. 813 Ma and ca. 720 Ma for dyke 1 and dyke 2, respectively; Figure 5). The near-concordance of the data (99.5–96.8%) results in a long projection to the lower intercepts, and though the associated ages thereby have relatively high uncertainties (±ca. 160 Myr), we interpret these to reflect the timing of minor Pb-loss associated with the affects of Pan-African disturbance. A similar result was obtained by *French and Heaman* [2010] in the NE Dharwar craton, where magmatic zircon and baddeleyite in a 2177 Ma E-W diabase dyke near Bandepalem are variably affected by a ca. 485 Ma Pb-loss event.

4.3. Paleomagnetism

[14] The mean direction of nine samples from the dated dyke near Karimnagar (dyke 1) is similar to the characteristic remanence directions obtained on most NE-trending dykes from this region [*Rao et al.*, 1990]. However, this direction (D = 63°, I =

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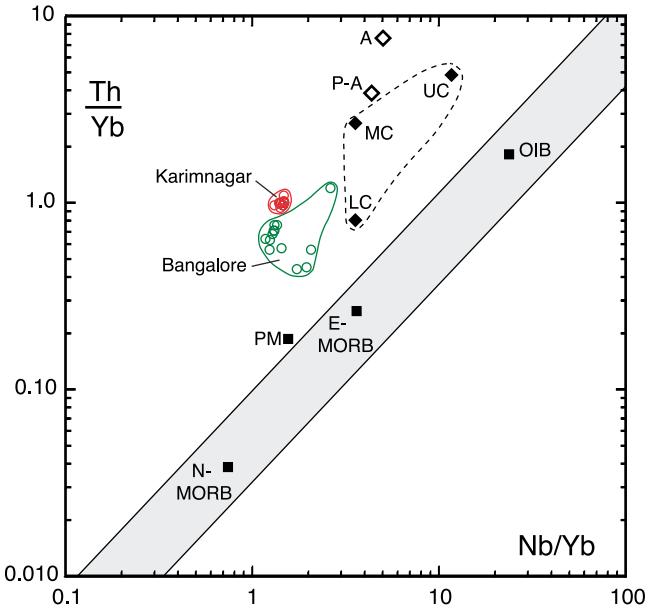


Figure 4. Immobile incompatible element (Th-Nb-Yb) ratio diagram for Karimnagar and Bangalore dykes, after *Pearce* [2008]. The steep dispersion of dyke compositions above the MORB-OIB array suggests significant crust-mantle interaction. Compositions for average felsic Archean (A) and post-Archean (P-A) crust, and for average lower (LC), middle (MC) and upper (UC) crust are taken from *Rudnick and Fountain* [1995]. Estimates of N-MORB, E-MORB, ocean island basalt (OIB) and Primordial Mantle (PM) are taken from *Sun and McDonough* [1989]. Shaded field represents MORB-OIB array, after *Pearce* [2008].

−6°) is in sharp contrast to the steep up inclinations found in the dated Penukonda, Chennekottapalle, Harohalli, and Henur dykes of the Bangalore swarm in the southern region [*Dawson and Hargraves*, 1994; *Halls et al.*, 2007b, and references therein] (Table 2). The observed difference in inclination, about 73° between the northern and southern sites, implies, if these remanences are primary, that dykes in the southern region were emplaced when the Dharwar craton was located at ∼69° south latitude whereas dykes in the northern region were at equatorial latitudes. Knowing continental drift patterns during the Phanerozoic (about 3 to 15 cm a^{-1}), it is unlikely that this craton could have drifted more than 6000 km (at more than 100 cm a^{-1}) during the time span of dyke emplacement, to account for their contrasting magnetization direction. However, two other dykes in the Karimnagar swarm were discovered to carry the characteristic steep up inclination direction (the dated Hyderabad dyke [dyke 2], and a second Karimnagar dyke [dyke 3]; Figures 1 and 6 and Table 2). This suspected primary remanence direction is preserved in the higher coercivity magnetic minerals (e.g., dykes 2 and 3 in Figure 6), and confirms our suspicion that most Karimnagar dykes are in fact overprinted by a secondary magnetization. This overprint extends into the region west of the Cuddapah basin where a ∼2.2 Ga dyke sampled along a length of 350 km shows a secondary magnetization with lower coercivity than that of the primary and a direction very close to that found by *Rao et al.* [1990] in the Karimnagar dykes (A. Kumar et al., manuscript in preparation, 2012).

5. Discussion

[15] The precise U-Pb geochronology presented here provides unambiguous evidence that the NE-trending Karimnagar swarm represents the northern extension of the Bangalore swarm, and thereby establishes the existence of a single geological entity (herein named the Dharwar giant dyke swarm) encompassing an area of 140,000 km^2 in the Dharwar craton, thus confirming the original conjecture by *Halls* [1982].

[16] A systematic progression in the strike of dykes within this swarm from north to south is illustrated by rose diagrams in Figure 1. In the north, the dated Karimnagar dyke (dyke 1) trends N48°E, while the dated Henur dyke at the southern end of the swarm has a strike direction of N110°E. However, all of the dykes with a SSE trend lie south of Latitude 12.5°N, either within or less than about 30 km from the southern granulite belt where major post-

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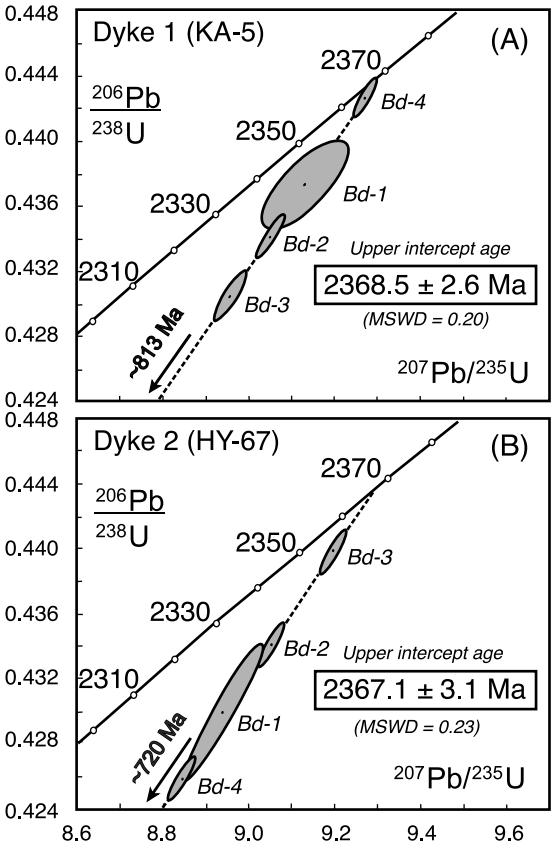


Figure 5. Concordia diagrams showing U-Pb ID-TIMS data for baddeleyite separated from Karimnagar dyke samples (a) dyke 1 (KA-5) and (b) dyke 2 (HY-67). Decay constants used are those of *Jaffey et al.* [1971]. Linear regression treatment and errors were calculated using the algorithms of *Ludwig* [2003]. Error ellipses and age uncertainties are shown at the 95% confidence level.

Archean faults occur (Figure 1). Several of these dykes (sites 13–15 in the work of *Halls et al.* [2007b]) have been completely overprinted and are both altered and sheared. Using Tables 1 and 3 of *Halls et al.* [2007b], and including only sites with N, number of dykes ≥4 and $\alpha_{95} \leq 15°$, the mean primary paleomagnetic direction of these dykes is $D_1 = 112°$, $I_1 = -77°$, $\alpha_{95} = 6°$, N = 7, and their mean trend is 295°. Bangalore dykes lying between Lat. 12.5°N and 13.5°N give $D_2 = 90°$, $I_2 = -83°$, $\alpha_{95} = 5°$, N = 9 and have a mean trend of 275°. Although the paleomagnetic directions are not significantly different at p = 0.05, the difference in mean trend (20°) is close to the difference in declination (22°). It is therefore possible that the ESE trending dykes have been rotated about vertical axes by later tectonism and thus are not included in assessing the total fan angle of the Dharwar swarm. The total fan angle is then about 40° (Figure 1) giving a focal point about 300 km west of the Indian Peninsula, a conclusion consistent with a previous speculation by *French and Heaman* [2010], which was based largely upon dyke trends alone, without geochronology, geochemical data or primary paleomagnetic evidence on the Karimnagar swarm itself.

[17] The curvature in the regional structural grain of the Archean basement together with the fanning pattern of the dykes raises the possibility that both have suffered a counterclockwise rotation of the northern part with respect to the southern. Using the same procedure as before, and including the results from both Karimnagar dykes 2 and 3 in Table 2, the mean paleomagnetic direction of all dykes north of Lat. 13.5°N is $D_3 = 80°$, $I_3 = -72°$, $\alpha_{95} = 6.5°$ and N = 6 with a mean dyke trend of 255°. Although this direction is not significantly different from D_2, I_2 at p = 0.05, there is nevertheless a continuing decrease in D as the dykes swing northward into more NE trends.

[18] A ~2.2 Ga, 350 km long, N-S dyke west of the Cuddapah basin curves in sympathy with the Archean structural grain and has been paleomagnetically sampled along its length (Kumar et al., manuscript in preparation, 2012). The extreme northern and southern sites show a significant (p = 0.05) difference in declination that supports the rotation, but the expected correlation between local dyke trend and remanence declination is less convincing. If rotation has occurred it is perhaps significant that the Cuddapah basin coincides with a region where rifting, increasing eastward, is required to produce the rotation. However, although 1.9 Ga mafic sills occur toward the base of the sedimentary sequence (U-Pb, baddeleyite: 1885.4 ± 3.1 Ma [*French et al.*, 2008]), gravity data do not support the presence of mafic rift products beneath the basin. The lack of significant deformation of the sedimentary sequence in the western part of the Cuddapah basin, suggests that the Dharwar craton has remained in its present form since at least 1.9 Ga. In addition, N-S trending dykes south of the Cuddapah basin and dated by U-Pb on baddeleyite at 2220 ± 4.9 Ma by *French and Heaman* [2010] may be the continuation of undated N-S trending dykes that lie on strike and to the north of the basin. We therefore conclude that without more data the curvature of the Archean structural grain must be assumed to be an original feature and the following discussion is based upon this premise.

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Table 1. U-Pb Analytical Results for Baddeleyite From the Karimnagar Dyke Swarm, Karimnagar-Hyderabad Region[a]

Sample Fraction	Description	Weight (μg)	U (ppm)	Th/U	Pb* (pg)	Pb_C (pg)	$^{206}Pb/^{204}Pb$	$^{206}Pb/^{238}U$	$\pm2\sigma$	$^{207}Pb/^{235}U$	$\pm2\sigma$	$^{207}Pb/^{206}Pb$	$\pm2\sigma$	$^{206}Pb/^{238}U$ Age (Ma)	$\pm2\sigma$	$^{207}Pb/^{235}U$ Age (Ma)	$\pm2\sigma$	$^{207}Pb/^{206}Pb$ Age (Ma)	$\pm2\sigma$	Disc. (%)	Corr. Coeff.
Dyke 1 – KA-5 Medium-Grained Diabase, WSW of Karimnagar; 18.21°N, 78.82°E																					
Bd-1	2 mbr, fl blades	0.2	115	0.11	10.2	1.1	589	0.437319	0.002218	9.12913	0.08095	0.151367	0.000912	2339.1	9.9	2351.0	8.1	2361.4	10.3	1.1	0.7567
Bd-2	3 m-dbr, fl blades and frags	0.4	128	0.16	28.6	0.5	3769	0.434222	0.001074	9.05000	0.02732	0.151159	0.000192	2324.7	4.8	2343.1	2.8	2359.1	2.2	1.7	0.9123
Bd-3	3 p-mbr, fl blades and frags, dull	0.35	101	0.11	22.1	0.6	2480	0.430574	0.001253	8.95467	0.03208	0.150666	0.000261	2308.3	5.6	2332.4	3.3	2353.5	3.0	2.3	0.8787
Bd-4	6 p-mbr euh fl blades	0.7	114	0.08	39.3	0.4	5017	0.442652	0.000964	9.26992	0.02299	0.151159	0.000192	2362.5	4.3	2365.1	2.3	2367.2	1.7	0.5	0.9124
Dyke 2 - HY-67 Medium-Grained Diabase, E of Hyderabad; 17.50°N, 78.87°E																					
Bd-1	1 pbr euh blade, dull	0.2	387	0.09	5.7	0.2	1563	0.429979	0.003473	8.93491	0.07704	0.151026	0.000356	2305.6	15.7	2331.4	7.9	2354.0	3.9	2.8	0.9622
Bd-2	3 p-mbr euh fl blades	0.5	406	0.08	25.3	0.4	3668	0.434157	0.001155	9.04969	0.02683	0.151179	0.000177	2324.4	5.2	2343.0	2.7	2359.3	2.0	1.7	0.9123
Bd-3	4 p-mbr euh fl blades	0.6	375	0.08	27.7	0.4	4541	0.439855	0.001164	9.19379	0.02682	0.151595	0.000186	2350.0	5.3	2357.5	2.7	2364.0	2.1	0.7	0.9163
Bd-4	6 pbr euh fl blades, dull	0.7	354	0.07	31.3	0.3	4365	0.426019	0.001115	8.84358	0.02624	0.150560	0.000173	2287.8	5.0	2322.0	2.6	2352.3	1.9	3.2	0.9247

[a]All analyzed fractions represent best quality baddeleyite grains, free of inclusions and cracks. Dull lustre implies variable zircon overgrowth. Abbreviations: pbr - pale brown (tan); mbr - medium brown; dbr - dark brown; euh - euhedral; fl - flat; frag - fragment. Fraction weights (in micrograms) are estimated from digitally measured grain dimensions and the density of baddeleyite (realistic errors are on the order of ±20%). Pb* is total amount (in picograms) of radiogenic Pb. Pb_C is total measured common Pb (in picograms) assuming the isotopic composition of laboratory blank: 206/204 - 18.221; 207/204 - 15.612; 208/204 - 39.360 (errors of 2%). Pb/U atomic ratios are corrected for spike, fractionation, blank, and, where necessary, initial common Pb; $^{206}Pb/^{206}Pb$ is corrected for spike and fractionation. Th/U is model value calculated from radiogenic $^{208}Pb/^{206}Pb$ ratio and $^{207}Pb/^{206}Pb$ age assuming concordance. Disc. (%) - per cent discordance for the given $^{207}Pb/^{206}Pb$ age.

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Table 2. Paleomagnetic Data for Dykes of the Karimnagar and Bangalore Swarms[a]

Dyke Swarm	N [n]	Mean, D (deg)	Mean, I (deg)	k	α_{95}, [A_{95}] (deg)	P_{lat} (°N)	P_{lon} (°E)	Dp (deg)	Dm (deg)
Suspected Primary Component									
Bangalore[b]	25	99.0	−79.4	83	[5.6]	15.7	56.9	–	–
Karimnagar - Dyke 3	1 [8]	68.9	−65.2	19.6	11.2	0.26	39.9	16.8	20.8
Karimnagar - Dyke 2[c]	1 [15]	102.8	−66.6	28.3	6.8	21.4	35.5	9.9	12.1
Suspected Secondary Component									
Karimnagar - Dyke 1[c]	1 [9]	63.4	−5.8	8.6	15.9	24.1	181	9.4	18.7
Karimnagar[d]	12	52.5	−24.5	21.8	8.7	31.9	197	–	–

[a]N, n: Number of dykes, samples; D, I: Magnetization declination, inclination. k: *Fisher* [1953] precision parameter. α_{95}, A_{95}: Radius of the 95% circle of confidence about the mean magnetization direction, and pole position. P_{lat}, P_{lon}: Latitude and longitude of mean pole position. Dp, Dm: Semi-minor and -major axes of the 95% confidence ellipse about the pole.
[b]Combined mean direction and pole for Bangalore dyke Component A of *Halls et al.* [2007b].
[c]Dykes dated, this study (U-Pb; baddeleyite).
[d]Mean direction and pole for Karimnagar dyke Component B of *Rao et al.* [1990].

[19] The spectacular radiating dyke pattern of the Dharwar swarm represents a configuration suggesting dyke propagation from a centrally located magma source (cf. 1.27 Ga Mackenzie dykes [*LeCheminant and Heaman*, 1989]). Moreover, its rapid emplacement (between 3 and 7 Myr, considering 2σ error limits) is characteristic of starting plume head eruptions such as that represented by the Mackenzie igneous event in Canada, and supports a model in which the Dharwar giant dyke swarm originated from a large mantle plume. Other features of plume head magmatism, such as coeval volcanic and plutonic rocks and associated uplift [*LeCheminant and Heaman*, 1989], are not known for the Dharwar giant dyke swarm, but these and age-equivalent diabase dyke swarms might be sought from Archean cratons worldwide, in order to determine the continental block(s) that may have existed west of India in Paleoproterozoic times.

[20] Paleomagnetic studies on the Dharwar giant dyke swarm (Table 2) suggest that, at 2.37 Ga, India was located at high (∼69°) southerly latitudes. The only other known Paleoproterozoic dyke swarm that was emplaced at comparably high southern latitudes (∼50°) is the 2410–2418 Ma Widgiemooltha swarm of western Australia [*Evans*, 1968; *Nemchin and Pidgeon*, 1998; *Doehler and Heaman*, 1998]. Possible continental configurations involving Australia and India at ∼2.4 Ga allow both dyke swarms to be part of a giant radiating pattern centered close to the south pole [*Halls et al.*, 2007b]. In contrast, ∼2.4 Ga paleomagnetic data from elsewhere suggest that other Archean continents were located at low latitudes during that time (e.g., Superior craton, based on poles from the 2.45 Ga Matachewan dykes in Canada).

[21] The trends of dykes in the Dharwar giant dyke swarm are everywhere approximately normal to the structural grain of the Archean host rocks. This feature is observed in several other Paleoproterozoic dyke swarms from Precambrian shield regions in Canada and Australia [*Halls*, 1978] and is also manifest in earlier Archean times - the NE-trending 3.5–3.2 Ga Barberton greenstone belt [e.g., *Armstrong et al.*, 1990] and the crosscutting NW-trending 2.97 Ga Baadplaas dyke swarm [*Olsson et al.*, 2010]. In each case the dykes intrude about 200 Myr after cratonization. The orthogonality has been explained as due to the persistence of a regional Late Archean stress field into the Paleoproterozoic, resulting in fracture propagation along the direction of maximum compression. However, a plume source for the Dharwar dykes may also produce a radiating pattern in maximum stress trajectories as shown by the Afar magmatic center in northern Africa (see stress map of *Heidbach et al.* [2007]). One conceivable explanation for the orthogonality is to assume a long-lived Archean plume (∼2.8–2.5 Ga) that ultimately creates, through subduction and successive accretion of island arcs or by collision of smaller continents (e.g., the western and eastern Dharwar blocks at ∼2.5 Ga), an antipodal supercontinent after the manner proposed by *Li and Zhong* [2009]. Breakup and dispersal of the supercontinent during a time period of about 2.5 to 2.35 Ga, by the growth of hot mantle beneath could then produce outward continental dispersal. Following *Torsvik et al.* [2008], plume or local "hot spot" centers, generated around the periphery of the positive thermal anomaly in the mantle, could initiate rift zones broadly tangential to the mantle thermal anomaly and produce "failed arm" dyke swarms that would propagate into the

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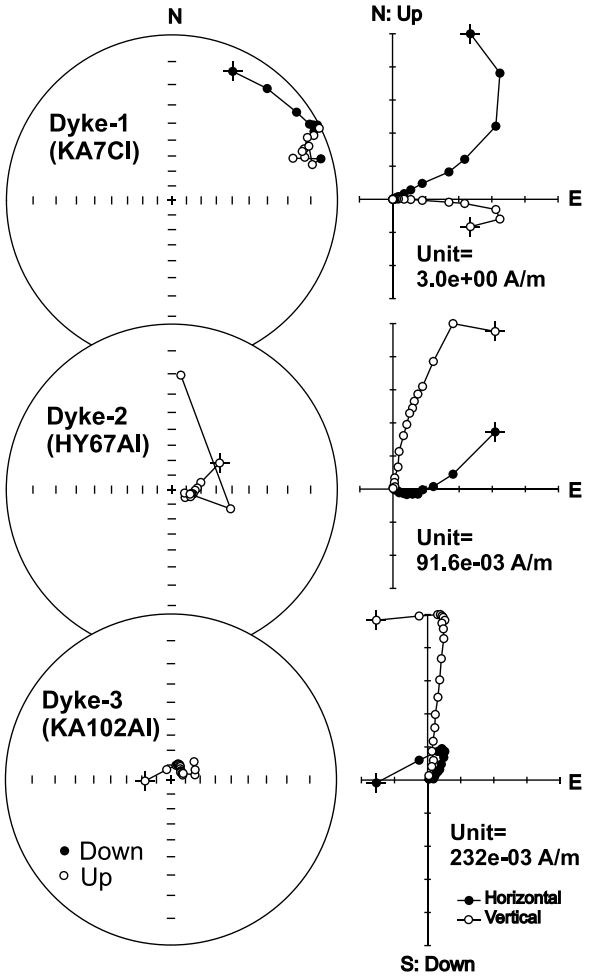


Figure 6. Stereoplots and vector diagrams, after AF demagnetization from 0 to 100 mT, for three Karimnagar dykes, two of which (dykes 1 and 2) have been dated and for which U-Pb results are shown in Figure 5. One sample, (dyke 1, KA-7) yields a NE-directed shallow inclination characteristic component, interpreted to be a secondary overprint, and a sample each from two dykes, (dyke 2, HY-67 and dyke 3, KA-102), reveals a higher coercivity component with a steep negative inclination, interpreted to be primary.

interior of the supercontinent and be approximately orthogonal to the pre-existing grain of granite-greenstone belts.

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